XINGDA LYU

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EDUCATION

University of Washington, Seattle *Expected Jun 2027*
Dual B.S. degrees in Statistics (Data Science Track) and Informatics (Honors)
Minor in Philosophy
Cumulative GPA: 3.9/4.0
Selected coursework: Statistical Machine Learning, Real Analysis, Data Structures and Algorithms, Mathematical Statistics & Probability I–III, Generalized Linear Models, Statistical Computing, Databases and Data Modeling.

PUBLICATIONS & MANUSCRIPTS

Lyu, Xingda*; Kaur, Kirandeep*; Zhou, Zhiyu; and Shah, Chirag, 2026. *Modeling Evolving Minds: Toward Epistemic Human-AI Collaboration.* Manuscript in preparation.

Lyu, Xingda*; Li, Stella*; et al., 2026. *On the User's Side: Benchmarking How Language Models Navigate Benefit–Enjoyment Conflicts.* Manuscript in preparation.

Lyu, Xingda et al., 2026. *The Missing Window: Evidence-Aware Forecasting for Long-Context Multivariate Time Series.* Manuscript in preparation.

*Equal contribution.

Kaur, Kirandeep; **Lyu, Xingda**; and Shah, Chirag, 2026. *Position: Knowing Isn't Understanding: Re-grounding Generative Proactivity with Epistemic and Behavioral Insight.*
Accepted at the International Conference on Machine Learning (ICML 2026). [paper]

Lyu, Xingda; Luo, Xinye; Lu, Honglin; and Yang, Shiqi, 2026. *PROS: A Proactive Multimodal Refinement Agent for Scientific Posters.*
Accepted and presented at the Workshop on Human-Interactive Generation and Editing (HiGen), CVPR 2026.

Lyu, Xingda; Lyu, Gongfu; Yan, Zitai; and Jiang, Yuxin, 2025. *P-RAG: Prompt-Enhanced Parametric RAG with LoRA and Selective CoT for Biomedical and Multi-Hop QA.*
In Proceedings of the International Conference on Computing and Data Science. [paper]

SELECTED PRESENTATIONS

Lyu, Xingda, 2026. *Decision-Making Under Unknown Unknowns in Proactive Agentic Systems.*
Lightning talk and poster presentation, ACM CHIIR Workshop on Human-Centered Proactive & Personalized Agents. [accepted papers]

Kaur, Kirandeep; Gupta, Vinayak; **Lyu, Xingda**; Gupta, Aditya, and Shah, Chirag, 2026. *Proactive Agents for Knowledge Gap Navigation.*
Poster presentation, Amazon Research Day 2026, AI Co-Scientist. [poster]

RESEARCH EXPERIENCE

Research Intern, Data Mining Group Summer 2026
Supervisor: Prof. Jiawei Han; PhD Mentor: Joey Chan University of Illinois Urbana-Champaign

- Selected for a Summer AI research program in the Data Mining Group, focused on machine learning, large language models, information retrieval, and AI agents.

- Conduct research in AI for science, particularly retrieval-augmented generation for retrosynthesis.

Research Assistant, InfoSeeking Lab Nov 2025 – Present
Supervisor: Prof. Chirag Shah; PhD Mentor: Kirandeep Kaur University of Washington

- Conduct research on proactive AI agents that support users under uncertainty, incomplete context, and unarticulated information needs.
- Co-authored an ICML 2026 position paper on epistemic and behavioral foundations for proactive agentic systems.
- Develop empirical extensions of this research direction through benchmark design, model-based experiments, and analysis of proactive agent behavior under epistemic uncertainty.

Research Collaborator, Proactive & Personalized LLM Assistants Apr 2026 – Present
PhD Collaborator: Stella Li University of Washington

- Co-develop a research project on calibration challenges in proactive and personalized LLM assistants.
- Study how assistants can provide helpful support while accounting for both explicit user requests and broader interaction context.
- Contribute to problem formulation, related work synthesis, prompt and example design, model output analysis, qualitative error analysis, and writing.

Research Project Lead, PROS Sep 2025 – Present
Independent Research Project University of Washington

- Led PROS, a self-directed research project on proactive multimodal refinement of scientific posters using editable PPTX representations.
- Led a small undergraduate team in building a source-grounded poster-editing pipeline combining structured PPTX representations, LLM/VLM-based diagnosis, and constraint-aware editing.
- Developed an executable taxonomy and staged repair framework for structural validity, layout reallocation, and scientific communication repair, achieving 82.0% macro diagnosis F1 and 73.0% average repair resolution.

Research Project Lead, P-RAG Nov 2024 – May 2025
Supervisor: Prof. David P. Woodruff Carnegie Mellon University

- Led P-RAG, a research project on retrieval-augmented biomedical and multi-hop question answering.
- Co-developed the method by integrating retrieval, LoRA-based parametric adaptation, selective chain-of-thought prompting, and data augmentation.
- Designed and ran experiments on PubMedQA and 2WikiMultihopQA, achieving 93.33% F1 on PubMedQA and improving total score on 2WikiMultihopQA from 17.83 to 33.44.
- Co-first-authored the resulting peer-reviewed conference paper.

PROFESSIONAL EXPERIENCE

Data Science Intern Jun 2025 – Aug 2025
Amazon Seattle, WA

- Built automated data-quality pipelines over 300K+ transportation records using Python and SQL.
- Designed routing optimization models projecting $5M+ in annual savings and 23% route utilization improvement.
- Built anomaly-detection analyses to identify refund gaps and underperforming carriers.

Data Science Intern Jul 2024 – Sep 2024
Baynovation San Jose, CA

- Built a refinance decision tool using real-time mortgage rate data.

- Built an interactive dashboard for mortgage scenario analysis and savings visualization.
- Created delinquency risk models using GLM, Random Forest, and XGBoost.

HONORS & SERVICE

UW iSchool Conference Travel Funding Award, to present accepted work at ICML 2026 *2026*

Student Volunteer, International Conference on Machine Learning (ICML) *2026*

Co-inventor on three CNIPA-granted utility patents for automated veterinary pharmaceutical production systems *2022–2023*

Co-inventor on a Chinese invention patent for a microbial agent for preventing wheat powdery mildew *2022*

SKILLS

Languages	Python, R, SQL, Java, JavaScript, HTML/CSS
ML / AI	PyTorch, Hugging Face, scikit-learn, LoRA/PEFT, RAG, LLM fine-tuning, multimodal modeling
Research / Evaluation	benchmark design, prompt design, LLM evaluation, ablation studies, statistical modeling, error analysis, qualitative evaluation
Tools	Git/GitHub, Linux, Jupyter Notebook, Flask